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Note on the U.S. Food and Drug Administration

The United States Food and Drug Administration (FDA) has evolved considerably since December, 1902, when Congress allowed the Bureau of Chemistry to feed several groups of young men a series of chemical-laden meals in order to assess the chemicals' toxicity. The men, dubbed the "Poison Squad" by the media, were volunteers from the Bureau of Chemistry in the U.S. Department of Agriculture, many of them chemists themselves, who wanted to support chief chemist Dr. Harvey Wiley's efforts to separate scientific facts on food safety from the inflammatory reports of mislabeled products, ineffective "cure-all" potions, and chemical preservatives in food at the turn of the century.

Wiley's "hygienic table studies" consisted of three wholesome meals containing potentially harmful substances. These substances were five chemicals commonly used as food preservatives at the time: borax, salicylic acid, sulfuric acid, sodium benzoate, and formaldehyde. Each preservative was fed to a different group of twelve men over the course of five years. Although the results were not surprising—the men suffered headaches, indigestion, and nausea but suffered no lasting effects—Wiley's experiment was highly publicized and raised an awareness of chemical additives in foods.[1]

This awareness and ensuing controversy led to the Pure Food and Drug Act of 1906, which laid the groundwork for today's FDA: a science-based law enforcement agency. The FDA included a Commissioner of Food and Drugs appointed by the President, who executed the Administration's responsibilities in research, law enforcement, and public education concerning food and drug-related issues. See **Exhibit 1** for a biography of 2007 Commissioner, Andrew von Eschenbach.

Although the FDA was heavily involved in new drug development, as all medical drugs had to be approved before they were allowed into the consumer market, the FDA did not develop or test products itself. Rather, the organization ensured that products were safe and that they met specific performance standards. The FDA also oversaw the product labeling for drugs and medical devices used by health professionals to ensure that the labels contained the information needed for proper use. In general, the product families that the FDA governed were checked for: safety, accurate labeling, safe manufacturing and handling, and potential new risks associated with the product's use in the consumer market.[2]

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[1] U.S. Food and Drug Administration, "The 'Poison Squad' and the Advent of Food and Drug Regulation," U.S. Food and Drug Administration website, http://www.fda.gov/FDAC/features/2002/602_squad.html, accessed January 22, 2007.

[2] U.S. Food and Drug Administration, "What FDA Regulates," U.S. Food and Drug Administration website, http://www.fda.gov/comments/regs/html, accessed January 22, 2007.

Professors Robert F. Higgins and Richard G. Hamermesh and Research Associate Virginia A. Fuller, Global Research Group, prepared this note as the basis for class discussion.

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History

While the Pure Food and Drug Act of 1906 enforced accurate and truthful labeling of products, product *safety* was still not regulated. This became tragically apparent in 1937 when 107 people, most of them children, died from toxic substances in Elixir of Sulfanilamide. Sulfanilamide was touted as a "wonder drug" that treated bacterial infections such as strep throat and gonorrhea, yet its liquid formulation—the one marketed for children—contained a poisonous chemical also found in antifreeze. Congress responded by enacting the Federal Food, Drug, and Cosmetic Act of 1938. This Act required that drug manufacturers must prove their products' safety before they could be marketed to the public.[3]

Over the course of the twentieth century, many government Acts further defined the mission of the FDA, which was called the Bureau of Chemistry until 1930. See **Exhibit 2** for a timeline of milestones in the development of the FDA. The Food and Drug Administration Act of 1988 officially established the FDA as an agency of the Department of Health and Human Services (as opposed to the Department of Agriculture). The 1997 the Food and Drug Administration Modernization Act imposed some of the most wide-ranging reforms in agency practices since 1938. Provisions of the act included: measures to accelerate the review of devices; regulation of the advertising of unapproved uses of approved drugs and devices; and, regulation of health claims for foods.

By 2006, the value of all products regulated by the FDA totaled $1 trillion, representing 25% of U.S. consumer spending. The FDA's 2006 budget was $1.6 billion, which amounted to $4 per year per taxpayer. The organization included 10,000 employees who were biologists, chemists, physicians, safety officers, biomedical engineers, pharmacologists, veterinarians, toxicologists, and specialists in public health education and communication. FDA staff visited approximately 16,000 sites per year to observe manufacturing practices and test products for contamination.[4]

FDA Responsibilities and Organization

The FDA's overarching mission—to promote and protect public health—was supported by the following tenets as set forth in the 1997 Act:

- Promote public health by promptly and efficiently reviewing clinical research and taking appropriate action on the marketing of regulated products in a timely manner.
- Protect public health by ensuring: a) that foods were safe, wholesome, sanitary, and properly labeled; b) that human and veterinary drugs were safe and effective; c) that there was reasonable assurance of the safety and effectiveness of devices intended for human use; d) that cosmetics were safe and properly labeled; and e) that public health and safety were protected from electronic product radiation.[5]

With its vast array of products to be reviewed, the FDA's regulatory approaches were as varied as the products it regulated. In order to fulfill its vast scope of responsibilities, the FDA was organized

[3] U.S. Food and Drug Administration, "The 1938 Food, Drug, and Cosmetic Act," U.S. Food and Drug Administration website, http://www.fda.gov/oc/history/historyoffda/section2.html, accessed January 22, 2007.

[4] U.S. Food and Drug Administration, "History of the FDA," U.S. Food and Drug Administration website, http://www.fda.gov/oc/history/historyoffda/default.htm, accessed January 22, 2007.

[5] U.S. Food and Drug Administration, "Section 903 of the Federal Food, Drug, and Cosmetic Act," U.S. Food and Drug Administration website, http://www.fda.gov/oc/fdama/fdamapln/sec903.htm, accessed January 22, 2007.

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into six Centers, with each handling the regulation of a different product type. See **Exhibit 3** for the FDA organizational chart.

The Centers

The Center for Devices and Radiological Health (CDRH) CDRH reviewed medical devices such as stents, sutures, or syringes; in short, anything that had contact with or went inside of the body. This included very simple items like tongue depressors and thermometers to very complex technologies such as heart pacemakers and dialysis machines. The FDA's role in electronic products covered the investigation and monitoring of radiation emitted from products such as microwave ovens, cell phones, x-ray equipment, lasers, and medical ultrasound and MRI machines.[6]

The National Center for Toxicological Research (NCTR) NCTR studied the biological mechanisms of action underlying the toxicity of products regulated by the FDA. This center also developed new methods of assessing human exposure to and risk of toxicity. [7]

The Center for Biologic Evaluation and Research (CBER) CBER reviewed products derived from living sources such as humans, animals, plants, or microorganisms. These products included blood, cellular and gene therapies, and vaccines. Although CBER was the primary review office for biologic products, some biologics therapies fell under the same review process as drugs and were therefore subject to the same examinations. As of October 1, 2003, the Center for Drug Evaluation and Research (CDER) began reviewing biologic therapy product applications such as antibodies and proteins, while CBER maintained jurisdiction over cellular products, vaccines, blood products, and gene therapies. See **Exhibit 4** for products affected by this transfer.[8]

The Center for Food Safety and Applied Nutrition (CFSAN) CFSAN monitored food products, dietary supplements, and cosmetics. The FDA safeguarded the nation's food supply by making sure that all ingredients used in foods were safe, and that food was free of contaminants such as disease-causing organisms, chemicals, or other harmful substances. The agency also reviewed new food additives before they were approved for use in foods. Additionally, the FDA monitored dietary supplements and the content of infant formulas. The FDA monitored cosmetic products to be sure that they were safe and properly labeled; however, cosmetics did not require premarket approval from the FDA.[9]

The Center for Veterinary Medicine (CVM) CVM regulated the manufacture and distribution of food additives and drugs given to animals, including animals from which human foods were derived, as well as food additives and drugs for pets. Before manufacturers were able to market animal drugs (including drugs used in animal feeds), they had to gain FDA approval by providing proof of their safety and effectiveness. Veterinary medical devices did not have to be pre-approved by FDA before marketing, but were still subject to FDA inspections once in use.

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[6] U.S. Food and Drug Administration, "Better Health Care with Quality Medical Devices: FDA on the Cutting Edge of Device Technology," U.S. Food and Drug Administration website, http://www.fda.gov/opacom/factsheets/justthefacts/5cdrh.html, accessed January 22, 2007.

[7] U.S. Food and Drug Administration, "NCTR's Mission," U.S. Food and Drug Administration website, http://www.fda.gov/nctr/overview/mission.htm, accessed January 22, 2007.

[8] U.S. Food and Drug Administration, "Transfer of Therapeutic Products to the Center for Drug Evaluation and Research," U.S. Food and Drug Administration website, http://www.fda.gov/cber/transfer/transfer.htm#1#1, accessed January 22, 2007.

[9] U.S. Food and Drug Administration, "Keeping the Nation's Food Supply Safe: FDA's Big Job Done Well," U.S. Food and Drug Administration website, http://www.fda.gov/opacom/factsheets/justthefacts/2cfsan.html, accessed January 22, 2007.

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Livestock drugs were evaluated for their safety to the environment or the animal. The FDA also ensured that any drug residues remaining in the animal once it was slaughtered for food were not harmful to consumers.[10]

The Center for Drug Evaluation and Research (CDER) CDER was the largest center, with about 2,000 employees. This center regulated all human drugs and biological products that were not regulated by CBER. (Refer to **Exhibit 4**.)

Both drugs and devices often had more than one purpose or action. To avoid confusion, a product was categorized as either drug or device based on its *primary* mode of action. For example, even though a drug-eluting stent contained chemical drugs, it was regulated by CDRH as a medical device because its primary action was to unblock an artery by mechanically stretching it open. Its secondary action was to release a chemical that prevented scar tissue from proliferating inside the artery. In contrast, a drug-eluting disk—a surgically implanted disk containing chemotherapeutic agents used to treat brain tumors—was regulated as a drug. Its primary mode of action was the chemotherapy's interaction with the tumor. The secondary mode of action was the continuous release of the drug by the device. Thus, drug-eluting disks were regulated as drugs.[11]

New Product Approval Process

Whether a product was a drug, device, biologic compound or any combination thereof, it had to undergo a series of FDA-monitored tests to ensure that it was safe and effective. The lifecycle of a new drug, biologic, or device followed the same basic development path. (See **Exhibit 5**.) There were three phases of clinical (human) trials a drug or biologic had to undergo before receiving market approval. Before that, however, the sponsoring company had to do its own research in order to convince the FDA that its product was worth testing on humans.

Preclinical Development

When developing a new drug or device, a company first estimated the cost of research and the potential for success in a) FDA approval and b) sales projections that would both recover costs and turn a profit. Upon agreeing to go ahead with development, company representatives decided on a regulatory strategy, including identifying which FDA Center possessed jurisdiction over their product and deciding how the product should be categorized.

Preclinical development was the stage in which the product was tested for manufacturability, storage ability, and safety in animal models. In the preclinical phase, the sponsor wanted to accrue enough evidence to show that there was both reasonable cause—in that the product was effective enough—and manageable risk—in that the product was safe enough—to allow the company to proceed with human studies in accordance with FDA guidelines.

Beyond this point, the different centers responsible for their respective product types took over and the development processes began to differ, only slightly, for the different product types.

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[10] U.S. Food and Drug Administration, "Safeguarding Animal Health to Protect Consumers," U.S. Food and Drug Administration website, http://www.fda.gov/opacom/factsheets/justthefacts/6cvm.html, accessed January 22, 2007.

[11] U.S. Food and Drug Administration, "Practical Issues in Combination Products," U.S. Food and Drug Administration website, http://www.fda.gov/oc/combination/presentations/advamed_files/outline/index.html, accessed January 22, 2007.

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CDER and Drug Development

The FDA's regulation of drugs, handled by CDER, was one of the organization's largest responsibilities. In order to be approved for public consumption, drugs had to meet FDA's standards for safety and efficacy. In the interest of getting their products to market, drug companies—called sponsors—worked closely with the FDA throughout their product's development and manufacturing stages to ensure that all FDA guidelines were followed. This reduced the chance of running into an unanticipated obstacle that would lengthen a new drug's time-to-market. See **Exhibit 6** for a description of the drug discovery, development, and review process.

Investigational New Drug (IND) Application The IND was a company's first official filing with CDER. This usually followed CDER's Pre-IND Consultation Program—a period of early communications between the sponsor and CDER during which CDER offered guidance on the IND submission process. The IND was the sponsoring company's opportunity to show the FDA that its drug was safe enough and viable enough to test in humans, based on its preclinical data in the following areas: Animal Pharmacology and Toxicology studies, Manufacturing Information, and Clinical Protocols and Investigator Information. Clinical trials were divided into three phases: Phase 1, Phase 2, and Phase 3, and the IND provided a detailed outline of the Phase 1 testing protocol. This included the number of subjects, dosing plan, necessary monitoring of vital signs, toxicity-based stoppage, and dosage adjustment rules. It also showed the ingredients of the compound, detailed the manufacturing process, and informed the FDA about the mechanics of the study including the control group.[12]

The IND was not an application for market approval; this would come much later. Rather, the IND allowed the sponsor to distribute an unapproved new chemical entity (NCE)—a therapeutic molecule or compound that had not been previously tested on humans—across state lines.[13] The ability to transport the drug was important to sponsors because they had to send it to many different clinical investigators at various testing sites around the country. (Human testing did occur outside the United States as well, and these trials were often accepted by the FDA.) After undergoing a review process, the IND was either accepted or rejected by the FDA. (See **Exhibit 7**.) If the IND was approved, the drug moved on to Phase 1 testing.

Phase 1 In Phase 1 trials, the drug was tested in a small number of healthy volunteers to monitor its absorption, metabolism, and toxicity in the body. Based on the results, the sponsoring company gained information about the safety and dosage of the drug. About 30% of new compounds in Phase 1 testing passed on to Phase 2.[14]

Phase 2 Phase 2 testing was done with a patient population that already had the condition or disease for which the new drug was designed and typically took place in a hospital or university research center. Phase 2 testing was designed to determine if the proposed drug produced the desired results and exemplified the drug's efficacy in its intended users. The drug's predicted or desired results were defined before the Phase 2 study as "endpoints." Physicians were responsible for each of the trial's study sites and were challenged throughout this phase to look for the most effective way of administering the drug to patients. The survival of a small drug company often rested on the outcome of Phase 2 tests, as admittance to Phase 3 was predicated on successful Phase 2 results. In

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[12] U.S. Food and Drug Administration, "Investigational New Drug (IND) Application Process," U.S. Food and Drug Administration website, http://www.fda.gov/cder/regulatory/applications/ind_page_1.htm, accessed January 22, 2007.

[13] Ibid.

[14] Joseph A. DiMasi, Ronald W. Hansen, and Henry G. Grabowski, "The Price of Innovation: New Estimates of Drug Costs," *Journal of Health Economics* 22 (2003): 151-185.

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2006, analysts estimated Phase 2 costs to be about $40 million over the course of two to four years. 50 percent of products tested in Phase 2 did not proceed to Phase 3 in 2004.[15]

End of Phase 2 Meeting Upon the successful completion of a Phase 2 study, company sponsors and CDER staff had the opportunity to meet and map out the final studies required for approval. This meeting, called the "End of Phase 2 Meeting," determined the scope and safety of Phase 3 trials, evaluated the plan and protocol for Phase 3, and identified additional information to support the drug's market application. The meeting resulted in an agreement between the FDA and the sponsor regarding what tasks had to be completed and what endpoints would be reached in order to secure market approval.[16]

Phase 3 Phase 3 was the most expensive phase of testing, as it lasted anywhere from two to three years, involved several hundred to several thousand patients, and cost between $50 million and $300 million.[17] Phase 3 was intended to provide further evidence of safety and efficacy under a wider variety of circumstances and subjects, including children or elderly patients. As the sponsor neared completion of Phase 3, it submitted a New Drug Application (NDA) to the FDA for review. (See **Exhibit 8**.)

NDA and Market Approval A drug sponsor's NDA formally appealed to the FDA for approval of a new product for sale and marketing in the United States. The FDA's review of an NDA could take as long as two to three years, and was undertaken by special review committees. Data gathered during animal studies and all phases of human trials for the product became part of the application, and the documents often contained hundreds of thousands of pages of information. The main purpose of the NDA was to provide the CDER review committees with enough information to decide on the following:

- Whether the drug was safe and effective in its proposed use(s), and whether the benefits of the drug outweighed the risks.
- Whether the drug's proposed labeling (package insert) was appropriate, and what information it should contain.
- Whether the methods used in manufacturing the drug and the controls used to maintain the drug's quality were adequate to preserve the drug's identity, strength, quality, and purity.

In informing the FDA's decision, the NDA outlined the complete history of the drug, from conception to Phase 3 clinical testing. This included the composition of the drug, the various tests performed on it, the drug's behavior in the body, and how it was going to be manufactured, stored, and distributed by the company.[18]

Once a decision was reached by the reviewers and their supervisors, it had to be evaluated (and approved) by the director of the applicable drug review division or office. The director reviewed an "action package" that contained an action letter outlining the reviewers' recommendation and any

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[15] Ibid.

[16] U.S. Food and Drug Administration, "Investigational Sponsor/FDA Meetings (End of Phase 2)," U.S. Food and Drug Administration website, http://www.fda.gov/cder/handbook/phas2mtg.htm, accessed January 22, 2007.

[17] Joseph A. DiMasi, Ronald W. Hansen, and Henry G. Grabowski, "The Price of Innovation: New Estimates of Drug Costs," *Journal of Health Economics* 22 (2003): 151-185.

[18] U.S. Food and Drug Administration, "New Drug Application (NDA) Process," U.S. Food and Drug Administration website, http://www.fda.gov/cder/regulatory/applications/NDA.htm, accessed January 22, 2007.

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data, CDER reviews and memos, and other information supporting the decision. Once a director signed an approval action letter, the product could be immediately legally marketed in the United States.[19]

CBER and Biologics Development

CBER's review of new biological products was very similar to the drug approval process. Following initial laboratory and animal testing, a biological product was studied in Phase 1, 2, and 3 clinical trials in humans under an IND. If the data generated by the studies showed that the product was safe and effective for its intended use, the data were submitted to CBER as part of a biologics license application (BLA)—CBER's version of an NDA—for review and, ultimately, market approval. Once a BLA was approved, the product was still subject to official lot release.[20]

CDRH and Device Development

The Medical Device Amendments of 1976 to the Federal Food, Drug, and Cosmetic Act established three regulatory classes for medical devices: Class I, II, and III. The three classes were based on the degree of regulatory control necessary to ensure that various types of devices were safe and effective. All medical devices were subject to "general controls" such as being manufactured under Good Manufacturing Practices (GMP) and being labeled in accordance with the FDA's labeling regulations.

Class I devices (such as elastic bandages and hand-held surgical instruments) were subject to the least regulatory control since they presented minimal potential harm to users. Class I devices did not require FDA approval prior to being marketed and most were exempt from GMP regulation as well.

Class II devices (such as powered wheelchairs and infusion pumps) were also subject to "special controls" such as mandatory performance standards and postmarket surveillance. Manufacturers of these devices were also required to complete a premarket notification (510(k)) process in order to obtain FDA clearance to market these devices.

Class III devices (such as replacement heart valves and coronary stents) were those that supported or sustained human life and, consequently, they were the most strictly regulated. Manufacturers of these devices were required to complete a premarket approval (PMA) process in order to obtain FDA clearance prior to marketing. One exception was Class III devices that were equivalent to devices legally marketed before 1976; these devices were allowed to be marketed through the 510(k) process.

After conducting preclinical studies for Class II and Class III devices, sponsors had to submit an Investigational Device Exemption (IDE) to CDRH before commencing clinical trials. An IDE was CDRH's equivalent of CDER's IND.[21]

Investigational Device Exemption (IDE) An IDE allowed sponsors to pursue clinical studies to collect safety and effectiveness data to support PMA or 510(k) applications for their

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[19] U.S. Food and Drug Administration, "NDA Actions," U.S. Food and Drug Administration website, http://www.fda.gov/cder/handbook/ndaapprv.htm, accessed January 22, 2007.

[20] U.S. Food and Drug Administration, "CBER Frequently Asked Questions," U.S. Food and Drug Administration website, http://answers-cber.fda.gov/cgi-bin/answers_cber.cfg/php/enduser/std_alp.php, accessed January 22, 2007.

[21] U.S. Food and Drug Administration, "Device Advice," U.S. Food and Drug Administration website, http://www.fda.gov/cdrh/devadvice/3132.html, accessed January 22, 2007.

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devices. The filing of an IDE usually followed earlier contact with CDRH staff in "Pre-IDE" meetings. Like pre-IND meetings, pre-IDE meetings took the form of informal or formal conference calls or in-person meetings. CDRH encouraged these early interactions as it helped increase the sponsor's understanding of requirements and regulations and thus helped speed the regulatory process. In addition, sponsors could submit preliminary information as "pre-IDE" submissions. This allowed CDRH staff to provide guidance on the more complex parts of the IDE application such as clinical protocol design and preclinical test results.

Clinical studies were most often conducted to support a PMA, as very few 510(k)'s required clinical data for their approval. After an IDE was submitted, CDRH then had 30 days to approve, approve "with conditions," or disapprove the application. Prior to initiation of clinical studies, Class III devices required approval by the Institutional Review Board (IRB) of each institution participating in the investigation. The IRB submission included the investigational plan as well as reports of prior studies.[22]

Premarket Notification (510(k)) Section 510(k) of the Food, Drug and Cosmetic Act required device manufacturers to notify the FDA, at least 90 days in advance, of their intent to market a medical device. This "Premarket Notification" was known as "the 510(k)." Anyone intending to market a device for which a Premarket Approval (PMA) was *not* required, had to submit a 510(k) to the FDA. This allowed the FDA to determine whether the device was equivalent to any device already on the market. Indeed, devices that required a 510(k) but not a PMA were those that had similar functions to pre-existing devices. In their 510(k), sponsors had to compare their device to one or more similar legally marketed devices and support their claims of substantial equivalency. The legally marketed device to which equivalence was drawn was commonly known as the "predicate." [23]

Premarket Approval Application (PMA) A PMA was much like a New Drug Application (NDA) or Biologic License Application (BLA), in that it was a private license granted to the applicant, allowing the device to be marketed. The PMA applicant was usually the entity that owned the rights to the data and other information submitted in support of FDA approval. The applicant was often the inventor or manufacturer of the device. Other applicants included partnerships, corporations, associations, and scientific or academic establishments.

CDRH required a minimum of 180 days to review the PMA and make a decision about the device. Before approving or denying a PMA, the appropriate CDRH advisory committee reviewed the PMA at a public meeting and provided its recommendation on whether the submission should be approved. PMA approval was contingent upon valid scientific data providing evidence of the device's safety and efficacy for its intended use. The FDA then notified the applicant that the PMA had been either approved or denied, and a notice was published on the Internet. The notice provided the data on which the decision was based and allowed interested parties an opportunity to petition the FDA within 30 days for reconsideration of the decision.[24]

On October 26, 2002 the Medical Device User Fee and Modernization Act of 2002 was signed into law. This authorized the FDA to charge a fee for medical device product reviews. These fees applied

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[22] U.S. Food and Drug Administration, "IDE Overview," U.S. Food and Drug Administration website, http://www.fda.gov/cdrh/devadvice/ide/index.shtml, accessed January 22, 2007.

[23] U.S. Food and Drug Administration, "Premarket Notification 510(k)," U.S. Food and Drug Administration website, http://www.fda.gov/cdrh/devadvice/314.html, accessed January 22, 2007.

[24] U.S. Food and Drug Administration, "Device Advice PMA Overview," U.S. Food and Drug Administration website, http://www.fda.gov/cdrh/devadvice/pma/, accessed January 22, 2007.

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to PMAs, Product Development Protocols (PDPs), BLAs for certain medical devices reviewed by CBER, certain supplements, and Premarket Notifcation 510(k)s. Payment had to be received on or before the time the application was submitted. If the applicant did not pay all fees, the FDA considered the application incomplete and would not accept it. See **Exhibit 9** for 2007 fee schedule.

Continued Risk Management

Because clinical trials were based on relatively small populations, the FDA played an active role in monitoring reports of adverse events post-market launch. The FDA received more than 400,000 problem reports a year which it had to follow up with investigations.[25]

The post-launch safety surveillance and ongoing technical support of drugs were known as Phase IV trials. These trials helped glean additional information, including a product's risks and benefits.[26] Manufacturers of certain types of medical devices were also required to perform postmarket surveillance in order to identify potential problems early and minimize risk to patients.[27] The FDA also closely monitored the manufacturing process for quality. If an investigation revealed a problem, the FDA could take a number of actions including requiring a product recall, withdrawing approval, requiring labeling changes, or issuing warnings to the medical community.[28]

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[25] U.S. Food and Drug Administration, "Protecting Consumers, Promoting Public Health," U.S. Food and Drug Administration website, http://www.fda.gov/oc/opacom/fda101/fda101text.html, accessed January 22, 2007.

[26] U.S. Food and Drug Administration, "The CDER Review Process," U.S. Food and Drug Administration website, http://www.fda.gov/cder/audiences/iact/forum/200609_hixon.pdf, accessed January 22, 2007.

[27] U.S. Food and Drug Administration, "Postmarket Surveillance Studies," U.S. Food and Drug Administration website, http://www.fda.gov/cdrh/devadvice/352.html#introduction/, accessed January 22, 2007.

[28] U.S. Food and Drug Administration, "Protecting Consumers, Promoting Public Health," U.S. Food and Drug Administration website, http://www.fda.gov/oc/opacom/fda101/fda101text.html, accessed January 22, 2007.

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Exhibit 1 Biography of 2007 FDA Commissioner Andrew von Eschenbach

Andrew C. von Eschenbach, M.D., was sworn in as the 20th Commissioner of the U.S. Food and Drug Administration on December 13, 2006.

Formerly the Director of the National Cancer Institute (NCI), Dr. von Eschenbach was a nationally recognized urologic surgeon and oncologist. Prior to being appointed to lead the NCI in January 2002, Dr. von Eschenbach served as Executive Vice President and Chief Academic Officer of the University of Texas M.D. Anderson Cancer Center in Houston, leading a faculty of more than 1,000 cancer researchers and clinicians. At M.D. Anderson he also served as Vice President for Academic Affairs and held the Roy M. and Phyllis Gough Huffington Clinical Research Distinguished Chair in Urologic Oncology.

Dr. von Eschenbach, as founding director of the Prostate Cancer Research Program, was instrumental in fostering integrated research programs in the biology, epidemiology, prevention, and treatment of prostate cancer at M.D. Anderson where he also directed the Genitourinary Cancer Center. He joined M.D. Anderson as a urologic oncology fellow in 1976 and was invited to join the faculty the following year. In 1983, he was named chairman of the Department of Urology. Other positions held at M.D. Anderson include Consulting Professor of Cell Biology and Professor of Urology.

Dr. von Eschenbach, himself a cancer survivor, was president-elect of the American Cancer Society at the time of his appointment to the NCI. In addition, he has made significant contributions to the scientific literature—more than 200 articles, books, and book chapters. Dr. von Eschenbach has also served as an editorial board member of several leading journals and on several organizational boards.

Many influential organizations have recognized Dr. von Eschenbach for his leadership and accomplishments; among them the American Medical Writers Association, the American Urological Association, and the Uniformed Services University of Health Sciences. He also has been included in "The Best Doctors in America" publications. Included among his many honors were the 2003 Carpe Diem Award from the Lance Armstrong Foundation; the Achievement Award from the 100 Black Men of Metropolitan Houston for his significant contributions to prostate cancer programs in the African-American community, the Julie Rogers "Spirit of Love" Award for demonstrating unparalleled dedication, commitment, and spirit in the fight against cancer, and the American Radium Society Janeway Medal for outstanding contribution to cancer research and the care of cancer patients. In 2006, Time Magazine chose Dr. von Eschenbach as one of the 100 most influential people to shape the world.

A native of Philadelphia, Dr. von Eschenbach earned a B.S. from St. Joseph's University in Philadelphia in 1963 and his medical degree from Georgetown University School of Medicine in 1967. Dr. von Eschenbach completed his internship at Philadelphia General Hospital and a residency in urologic surgery at Pennsylvania Hospital in Philadelphia. He was then was an instructor in urology at the University of Pennsylvania School of Medicine. He also served as a Lieutenant Commander in the U.S. Navy Medical Corps.

Source: http://www.fda.gov/oc/voneschenbach/bio.html, accessed January 17, 2007.

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Exhibit 2 Timeline of FDA Milestones

1820 Eleven physicians meet in Washington, D.C., to establish the **U.S. Pharmacopeia**, the first compendium of standard drugs for the United States.

1848 **Drug Importation Act** passed by Congress requires U.S. Customs Service inspection to stop entry of adulterated drugs from overseas.

1862 **President Lincoln** appoints a chemist, Charles M. Wetherill, to serve in the new Department of Agriculture. This was the beginning of the Bureau of Chemistry, the predecessor of the Food and Drug Administration.

1883 **Dr. Harvey W. Wiley** becomes chief chemist, expanding the Bureau of Chemistry's food adulteration studies. Campaigning for a federal law, Dr. Wiley is called the "Crusading Chemist" and "Father of the Pure Food and Drugs Act." He retired from government service in 1912 and died in 1930.

1898 Association of Official Agricultural Chemists (now AOAC International) establishes a **Committee on Food Standards** headed by Dr. Wiley. States begin incorporating these standards into their food statutes.

1902 The **Biologics Control Act** is passed to ensure purity and safety of serums, vaccines, and similar products used to prevent or treat diseases in humans.

Congress appropriates $5,000 to the Bureau of Chemistry to study **chemical preservatives and colors** and their effects on digestion and health. Dr. Wiley's studies draw widespread attention to the problem of food adulteration. Public support for passage of a federal food and drug law grows.

1906 The original **Food and Drugs Act** is passed by Congress on June 30 and signed by President Theodore Roosevelt. It prohibits interstate commerce in misbranded and adulterated foods, drinks and drugs. The **Meat Inspection Act** is passed the same day.

1907 First **Certified Color Regulations**, requested by manufacturers and users, list seven colors found suitable for use in foods.

1913 **Gould Amendment** requires that food package contents be "plainly and conspicuously marked on the outside of the package in terms of weight, measure, or numerical count."

1927 The Bureau of Chemistry is reorganized into two separate entities. Regulatory functions are located in the **Food, Drug, and Insecticide Administration**, and nonregulatory research is located in the **Bureau of Chemistry and Soils**.

1930 **McNary-Mapes Amendment** authorizes FDA standards of quality and fill-of-container for canned food, excluding meat and milk products.

The name of the Food, Drug, and Insecticide Administration is shortened to Food and Drug Administration (FDA) under an agricultural appropriations act.

1933 FDA recommends a complete revision of the obsolete **1906 Food and Drugs Act**. The first bill is introduced into the Senate, launching a five-year legislative battle.

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1937 **Elixir of Sulfanilamide**, containing the poisonous solvent diethylene glycol, kills 107 persons, many of whom are children, dramatizing the need to establish drug safety before marketing and to enact the pending food and drug law.

1938 **The Federal Food, Drug, and Cosmetic (FDC) Act** of 1938 is passed by Congress.

1939 **First Food Standards** issued (canned tomatoes, tomato purée, and tomato paste).

1940 **FDA transferred** from the Department of Agriculture to the Federal Security Agency, with Walter G. Campbell appointed as the first Commissioner of Food and Drugs.

1941 **Insulin Amendment** requires FDA to test and certify purity and potency of this lifesaving drug for diabetes.

1945 **Penicillin Amendment** requires FDA testing and certification of safety and effectiveness of all penicillin products. Later amendments extended this requirement to all antibiotics. In 1983 such control was found no longer needed and was abolished.

1949 FDA publishes **guidance to industry** for the first time. This guidance, "Procedures for the Appraisal of the Toxicity of Chemicals in Food," came to be known as the "black book."

1950 In **Alberty Food Products Co. v. U.S.** , a court of appeals rules that the directions for use on a drug label must include the purpose for which the drug is offered. Therefore, a worthless remedy cannot escape the law by not stating the condition it is supposed to treat.

Delaney Committee starts congressional investigation of the safety of chemicals in foods and cosmetics, laying the foundation for the 1954 Miller Pesticide Amendment, the 1958 Food Additives Amendment, and the 1960 Color Additive Amendment.

1953 **Federal Security Agency** becomes the Department of Health, Education, and Welfare (HEW).

Factory Inspection Amendment requires FDA to give manufacturers written reports of conditions observed during inspections and analyses of factory samples.

1954 **Miller Pesticide Amendment** spells out procedures for setting safety limits for pesticide residues on raw agricultural commodities.

First large-scale **radiological examination of food** carried out by FDA when it received reports that tuna suspected of being radioactive was being imported from Japan following atomic blasts in the Pacific. FDA begins monitoring around the clock to meet the emergency.

1955 The **Division of Biologics Control** became an independent entity within the National Institutes of Health, after polio vaccine thought to have been inactivated is associated with about 260 cases of polio.

1958 **Food Additives Amendment** enacted, requiring manufacturers of new food additives to establish safety. The Delaney proviso prohibits the approval of any food additive shown to induce cancer in humans or animals.

FDA publishes in the Federal Register the first list of **substances generally recognized as safe (GRAS).** The list contains nearly 200 substances.

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1962 **Thalidomide**, a new sleeping pill, is found to have caused birth defects in thousands of babies born in Western Europe. News reports on the role of Dr. Frances Kelsey, FDA medical officer, in keeping the drug off the U.S. market, arouse public support for stronger drug regulation.

Consumer Bill of Rights is proclaimed by President John F. Kennedy in a message to Congress. Included are the right to safety, the right to be informed, the right to choose, and the right to be heard.

1966 FDA contracts with the National Academy of Sciences/National Research Council to evaluate the **effectiveness of 4,000 drugs** approved on the basis of safety alone between 1938 and 1962.

Fair Packaging and Labeling Act requires all consumer products in interstate commerce to be honestly and informatively labeled, with FDA enforcing provisions on foods, drugs, cosmetics, and medical devices.

1968 **FDA Bureau of Drug Abuse Control** and Treasury Department Bureau of Narcotics are transferred to the Department of Justice to form the Bureau of Narcotics and Dangerous Drugs (BNDD), consolidating efforts to police traffic in abused drugs.

1969 The **White House Conference on Food, Nutrition, and Health** recommends systematic review of GRAS substances in light of FDA's ban of the artificial sweetener cyclamate. President Nixon orders FDA to review its GRAS list.

1970 FDA requires the first **patient package insert:** oral contraceptives must contain information for the patient about specific risks and benefits.

Environmental Protection Agency established; takes over FDA program for setting pesticide tolerances.

1971 **National Center for Toxicological Research** is established in the biological facilities of the Pine Bluff Arsenal in Arkansas. Its mission is to examine biological effects of chemicals in the environment, extrapolating data from experimental animals to human health.

1972 **Over-the-Counter Drug Review** begun to enhance the safety, effectiveness and appropriate labeling of drugs sold without prescription.

Regulation of Biologics—including serums, vaccines, and blood products—is transferred from NIH to FDA.

1973 **THE U.S. Supreme Court** upholds the 1962 drug effectiveness law and endorses FDA action to control entire classes of products by regulations rather than to rely only on time-consuming litigation.

1976 **Medical Device Amendments** passed to ensure safety and effectiveness of medical devices, including diagnostic products. The amendments require manufacturers to register with FDA and follow quality control procedures. Some products must have pre-market approval by FDA; others must meet performance standards before marketing.

Vitamins and Minerals Amendments ("Proxmire Amendments") stop FDA from establishing standards limiting potency of vitamins and minerals in food supplements or regulating them as drugs based solely on potency.

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1977 **Saccharin Study and Labeling Act** passed by Congress to stop FDA from banning the chemical sweetener but requiring a label warning that it has been found to cause cancer in laboratory animals.

1982 **Tamper-resistant Packing Regulations** issued by FDA to prevent poisonings such as deaths from cyanide placed in Tylenol capsules. The Federal Anti-Tampering Act passed in 1983 makes it a crime to tamper with packaged consumer products.

1983 **Orphan Drug Act** passed, enabling FDA to promote research and marketing of drugs needed for treating rare diseases.

1984 **Drug Price Competition and Patent Term Restoration Act** expedites the availability of less costly generic drugs by permitting FDA to approve applications to market generic versions of brand-name drugs without repeating the research done to prove them safe and effective.

1985 **AIDS test for blood** approved by FDA in its first major action to protect patients from infected donors.

1987 **Investigational drug regulations revised** to expand access to experimental drugs for patients with serious diseases with no alternative therapies.

1988 **Food and Drug Administration Act** of 1988 officially establishes FDA as an agency of the Department of Health and Human Services with a Commissioner of Food and Drugs appointed by the President with the advice and consent of the Senate, and broadly spells out the responsibilities of the Secretary and the Commissioner for research, enforcement, education, and information.

The **Prescription Drug Marketing Act** bans the diversion of prescription drugs from legitimate commercial channels. Congress finds that the resale of such drugs leads to the distribution of mislabeled, adulterated, subpotent, and counterfeit drugs to the public. The new law requires drug wholesalers to be licensed by the states; restricts reimportation from other countries; and bans sale, trade or purchase of drug samples, and traffic or counterfeiting of redeemable drug coupons.

Generic Animal Drug and Patent Term Restoration Act extends to veterinary products benefits given to human drugs under the 1984 Drug Price Competition and Patent Term Restoration Act. Companies can produce and sell generic versions of animal drugs approved after October 1962 without duplicating research done to prove them safe and effective. The act also authorizes extension of animal drug patents.

1990 **Nutrition Labeling and Education Act** requires all packaged foods to bear nutrition labeling and all health claims for foods to be consistent with terms defined by the Secretary of Health and Human Services.

Safe Medical Devices Act is passed, requiring nursing homes, hospitals, and other facilities that use medical devices to report to FDA incidents that suggest that a medical device probably caused or contributed to the death, serious illness, or serious injury of a patient. Manufacturers are required to conduct post-market surveillance on permanently implanted devices whose failure might cause serious harm or death, and to establish methods for tracing and locating patients depending on such devices. The act authorizes FDA to order device product recalls and other actions.

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1991 Regulations published to **Accelerate the Review of Drugs** for life-threatening diseases.

1992 **Prescription Drug User Fee Act** requires drug and biologics manufacturers to pay fees for product applications and supplements, and other services. The act also requires FDA to use these funds to hire more reviewers to assess applications.

Nutrition facts, basic per-serving nutritional information, are required on foods under the Nutrition Labeling and Education Act of 1990. Based on the latest public health recommendations, FDA and the Food Safety and Inspection Service of the Department of Agriculture recreate the food label to list the most important nutrients in an easy-to-follow format.

1994 **Dietary Supplement Health and Education Act** establishes specific labeling requirements, provides a regulatory framework, and authorizes FDA to promulgate good manufacturing practice regulations for dietary.

1995 FDA declares **cigarettes** to be "drug delivery devices." Restrictions are proposed on marketing and sales to reduce smoking by young people.

A series of **proposed reforms to reduce regulatory burden** on pharmaceutical manufacturers is announced, including an expansion of allowable promotional material on approved uses of drugs that firms can distribute to health professionals, streamlining certain elements in the documentation of investigational drug studies, and a reduction in both environmental impact filings and pre-approval requirements in tablet manufacture.

1997 **Food and Drug Administration Modernization Act** reauthorizes the Prescription Drug User Fee Act of 1992 and mandates the most wide-ranging reforms in agency practices since 1938. Provisions include measures to accelerate review of devices, regulate advertising of unapproved uses of approved drugs and devices, and regulate health claims for foods.

1998 FDA promulgates the **Pediatric Rule,** a regulation that requires manufacturers of selected new and extant drug and biological products to conduct studies to assess their safety and efficacy in children.

1999 **ClinicalTrials.gov** is founded to provide the public with updated information on enrollment in federally and privately supported clinical research, thereby expanding patient access to studies of promising therapies.

A final rule mandates that all over-the-counter drug labels must contain data in a standardized format. These **drug facts** are designed to provide the patient with easy-to-find information, analogous to the nutrition facts label for foods.

2000 The U.S. Supreme Court, upholding an earlier decision in Food and Drug Administration v. Brown & Williamson Tobacco Corp. et al., ruled 5-4 that **FDA does not have authority to regulate tobacco as a drug.** Within weeks of this ruling, FDA revokes its final rule, issued in 1996, that restricted the sale and distribution of cigarettes and smokeless tobacco products to children and adolescents, and that determined that cigarettes and smokeless tobacco products are combination products consisting of a drug (nicotine) and device components intended to deliver nicotine to the body.

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2002 In the wake of the events of September 11, 2001, the **Public Health Security and Bioterrorism Preparedness and Response Act of 2002** is designed to improve the country's ability to prevent and respond to public health emergencies, and provisions include a requirement that FDA issue regulations to enhance controls over imported and domestically produced commodities it regulates.

Under the **Medical Device User Fee and Modernization Act,** fees are assessed sponsors of medical device applications for evaluation, provisions are established for device establishment inspections by accredited third-parties, and new requirements emerge for reprocessed single-use devices.

The **Office of Combination Products** is formed within the Office of the Commissioner, as mandated under the Medical Device User Fee and Modernization Act, to oversee review of products that fall into multiple jurisdictions within FDA.

An effort to enhance and update the regulation of manufacturing processes and end-product quality of animal and human drugs and biological medicines is announced, the **current good manufacturing practice (cGMP) initiative.**

2003 **The Medicare Prescription Drug Improvement and Modernization Act requires,** among other elements, that a study be made of how current and emerging technologies can be utilized to make essential information about prescription drugs available to the blind and visually impaired.

The **Animal Drug User Fee Act** permits FDA to collect subsidies for the review of certain animal drug applications from sponsors, analogous to laws passed for the evaluation of other products FDA regulates, ensuring the safety and effectiveness of drugs for animals and the safety of animals used as foodstuffs.

2004 **Project BioShield Act of 2004** authorizes FDA to expedite its review procedures to enable rapid distribution of treatments as countermeasures to chemical, biological, and nuclear agents that may be used in a terrorist attack against the U. S., among other provisions.

Passage of the **Food Allergy Labeling and Consumer Protection Act** requires the labeling of any food that contains a protein derived from any one of the following foods that, as a group, account for the vast majority of food allergies: peanuts, soybeans, cow's milk, eggs, fish, crustacean shellfish, tree nuts, and wheat.

2005 Formation of the **Drug Safety Board** is announced, consisting of FDA staff and representatives from the National Institutes of Health and the Veterans Administration. The Board will advise the Director, Center for Drug Evaluation and Research, FDA, on drug safety issues.

Source: http://www.fda.gov/opacom/backgrounders/miles.html, accessed January 17, 2007.

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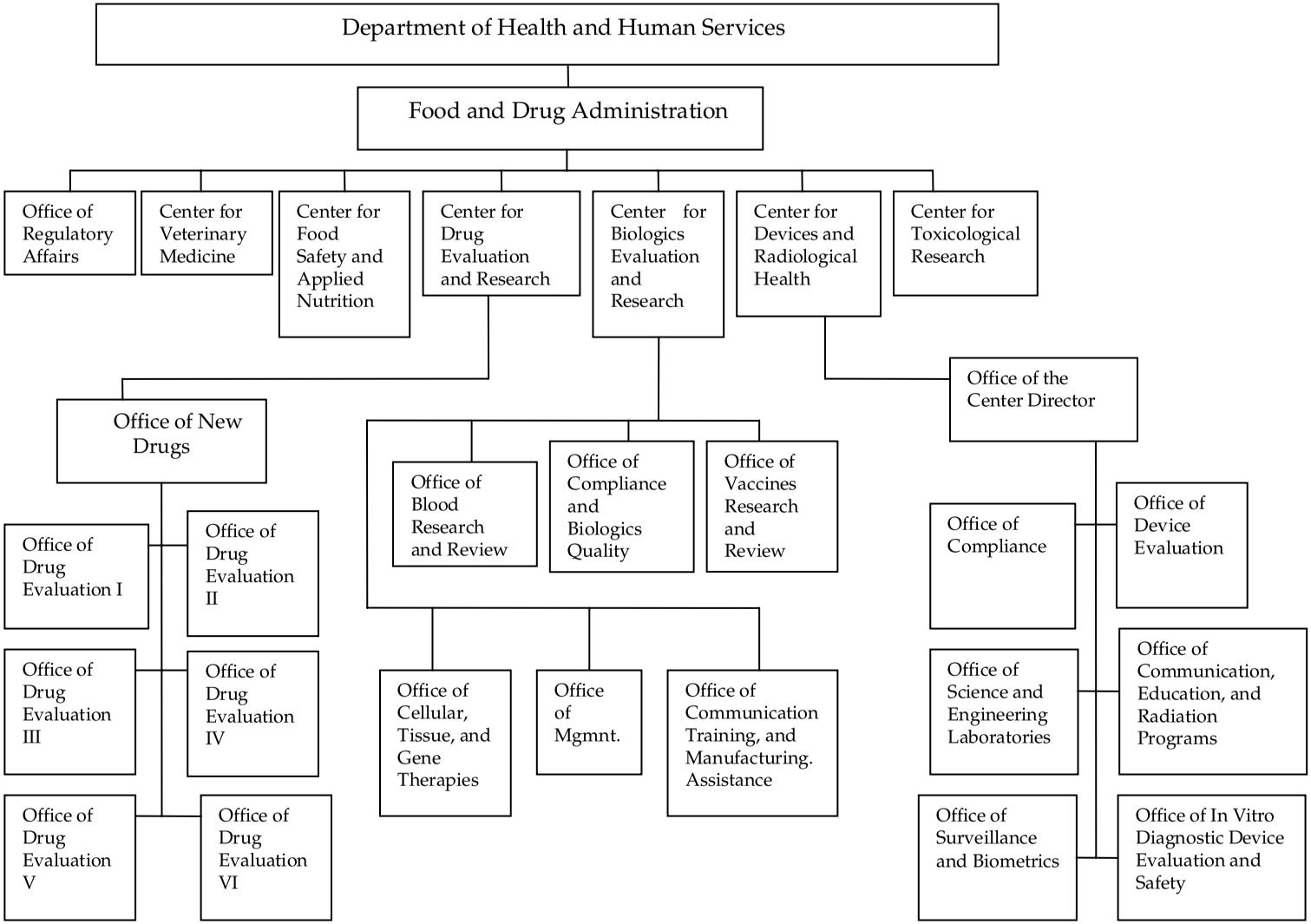
Exhibit 3 FDA Organizational Chart



Source: Adapted from U.S. Food and Drug Administration and casewriter research.

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Exhibit 4 Product Categories Affected by CBER-to-CDER Transfer

Categories of Therapeutic Biological Products Transferred to CDER

- Monoclonal antibodies for in vivo use.
- Proteins intended for therapeutic use, including cytokines (e.g. interferons), enzymes (e.g. thrombolytics), and other novel proteins, except for those that are specifically assigned to CBER (e.g., vaccines and blood products). This category includes therapeutic proteins derived from plants, animals, or microorganisms, and recombinant versions of these products.
- Immunomodulators (non-vaccine and non-allergenic products intended to treat disease by inhibiting or modifying a pre-existing immune response).
- Growth factors, cytokines, and monoclonal antibodies intended to mobilize, stimulate, decrease or otherwise alter the production of hematopoietic cells in vivo.

Categories of Therapeutic Biological Products Remaining in CBER

- Cellular products, including products composed of human, bacterial or animal cells (such as pancreatic islet cells for transplantation), or from physical parts of those cells (such as whole cells, cell fragments, or other components intended for use as preventative or therapeutic vaccines).
- Gene therapy products. Human gene therapy/gene transfer is the administration of nucleic acids, viruses, or genetically engineered microorganisms that mediate their effect by transcription and/or translation of the transferred genetic material, and/or by integrating into the host genome. Cells may be modified in these ways ex vivo for subsequent administration to the recipient, or altered in vivo by gene therapy products administered directly to the recipient.
- Vaccines (products intended to induce or increase an antigen specific immune response for prophylactic or therapeutic immunization, regardless of the composition or method of manufacture).
- Allergenic extracts used for the diagnosis and treatment of allergic diseases and allergen patch tests.
- Antitoxins, antivenins, and venoms.
- Blood, blood components, plasma derived products (for example, albumin, immunoglobulins, clotting factors, fibrin sealants, proteinase inhibitors), including recombinant and transgenic versions of plasma derivatives, (for example clotting factors), blood substitutes, plasma volume expanders, human or animal polyclonal antibody preparations including radiolabeled or conjugated forms, and certain fibrinolytics such as plasma-derived plasmin, and red cell reagents.

Source: http://www.fda.gov/cber/transfer/transfer.htm#1#1, accessed November 6, 2006.

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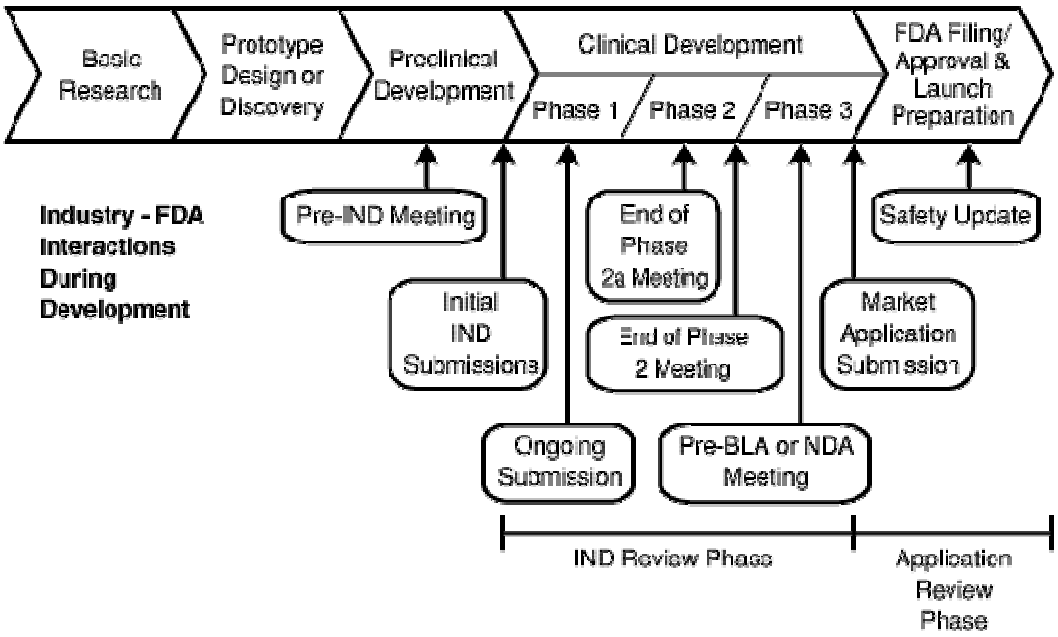
Exhibit 5 Industry—FDA Interaction During Drug Approval Process



Developers often met with the FDA before submitting an investigational new drug application (IND) to discuss early development plans. The IND had to be filed and cleared with the FDA before human testing could begin in the United States. During the clinical phase, there were ongoing submissions of new protocols and testing results. Developers often requested additional meetings to get FDA agreement on the methods proposed for evaluation of safety or efficacy, and also on manufacturing issues.

Prior to 1987, the FDA reviewers were kept independent from the sponsor companies. Unfortunately, this separation allowed many clinical studies to proceed even though their designs were not acceptable to FDA reviewers.

Source: http://www.fda.gov/oc/initiatives/criticalpath/whitepaper.html#execsummary, accessed November 6, 2006.

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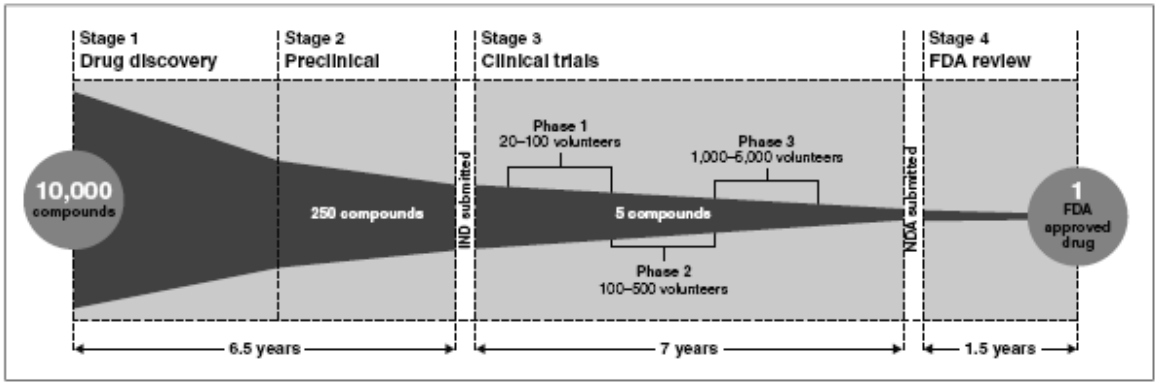
Exhibit 6 Drug Discovery, Development, and Review Process



Source: Pharmaceutical Research and Manufacturers of America, http://www.phrma.org/innovation/.

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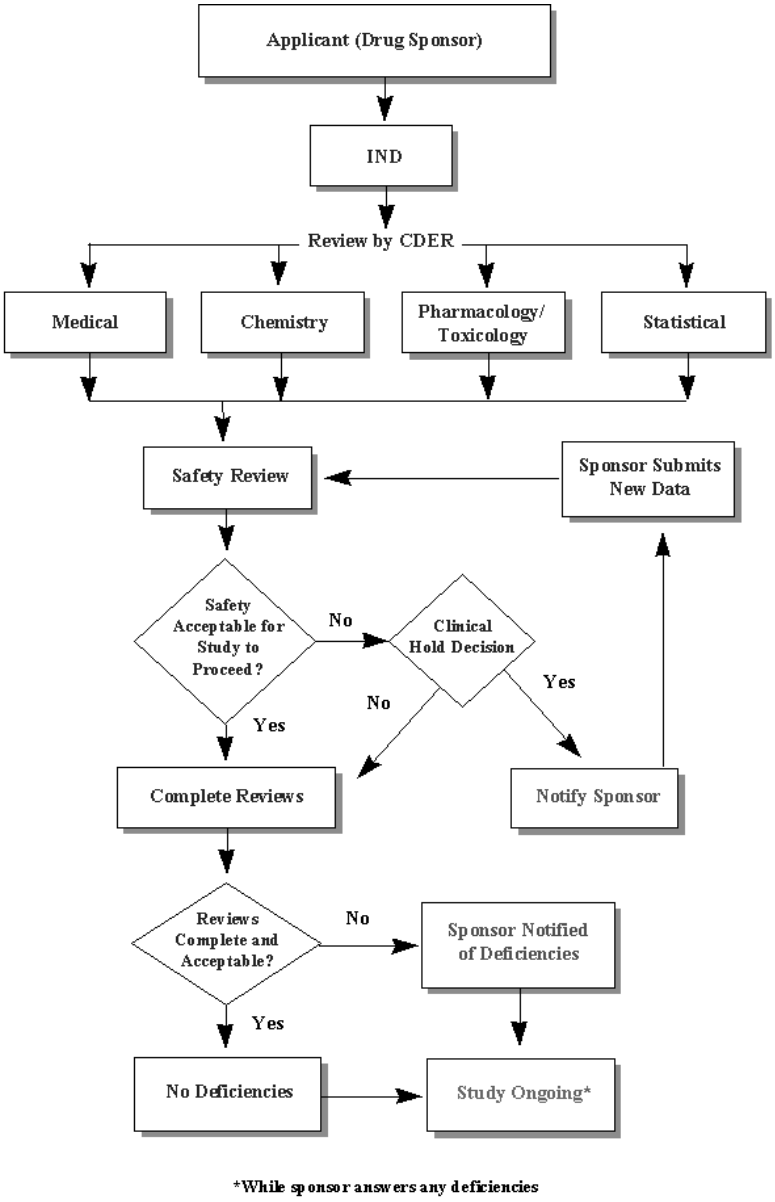
Exhibit 7 IND Approval Flowchart



"Study Ongoing" meant that clinical studies could be initiated upon expiry of CDER's 30-day review period, unless a clinical hold had been placed. If the sponsor was notified of deficiencies that were not serious enough to warrant a clinical hold, the sponsor would address these deficiencies in the subsequent phases of the study.

Source: http://www.fda.gov/cder/handbook/ind.htm, accessed November 20, 2006.

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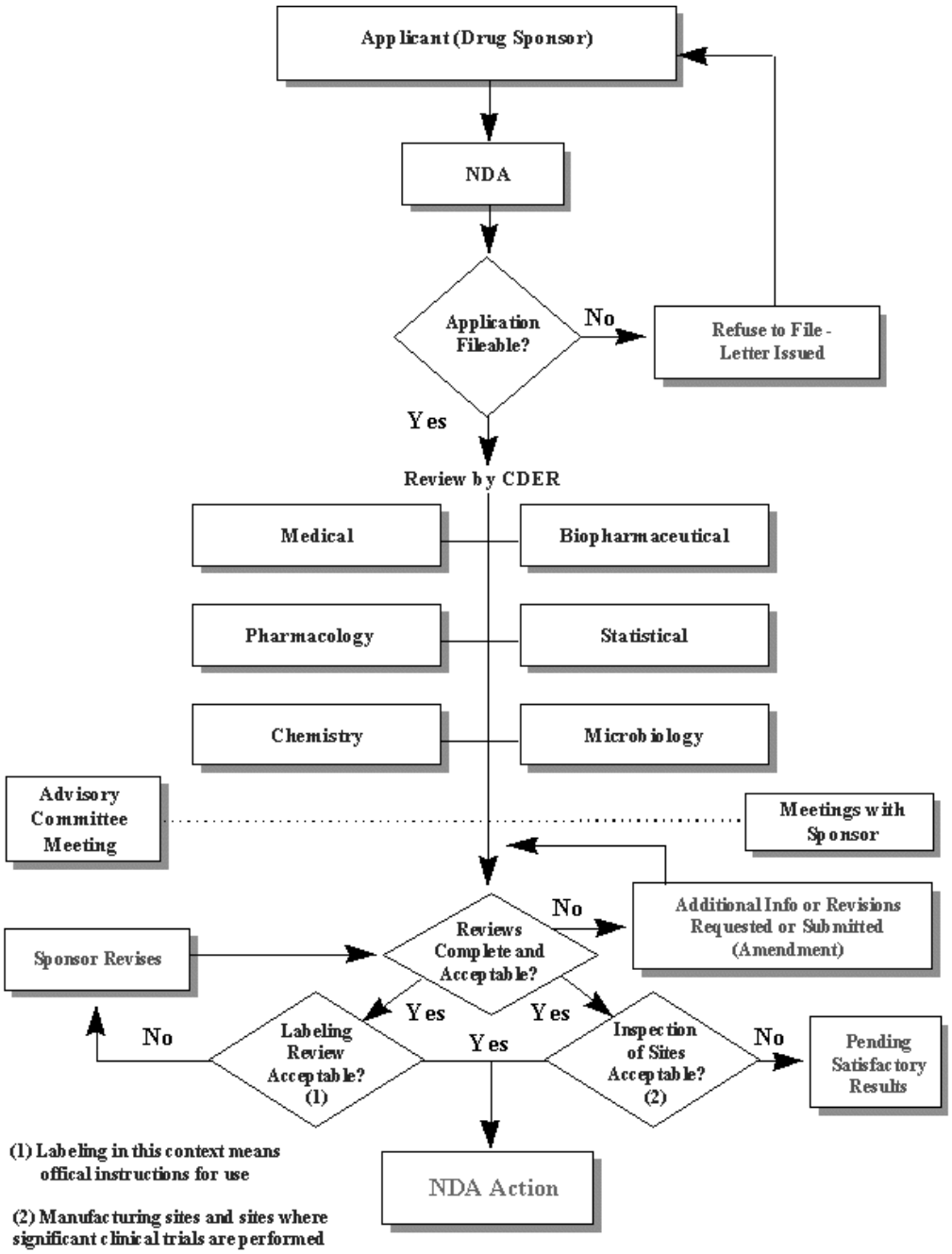
Exhibit 8 NDA/BLA Approval Process



Source: http://www.fda.gov/cder/handbook/nda.htm, accessed November 6, 2006.

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Exhibit 9 FY 2007 Device Review User Fees (U.S. Dollars)

FY 2007 Device Review User Fees (U.S. Dollars)		
Type of Application	**Standard Fee**	**Small Business Fee (<$100 million sales)**
Premarket Application (PMA, PDP, BLA, PMR)* **NOTE: The fee is waived for the *first* premarket application from firms with gross receipts or sales <$30 million.**	$281,600	$107,008
Premarket Report (PMR) - premarket approval application for a reprocessed device	$281,600	$107,008
Panel-track Supplement	$281,600	$107,008
Efficacy Supplement (for BLA)	$281,600	$107,008
180-day Supplement	$60,544	$23,007
Real-time Supplement	$20,275	$7,705

* PMA=Premarket Approval; PDP=Product Development Protocol; BLA=Biologics License Application; PMR=Premarket Report (for a reprocessed device)

Source: http://www.fda.gov/cdrh/devadvice/pma/userfees.html, accessed January 18, 2007.

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